VA Ann Arbor Guidance for Using the BioFire BCID2 Panel

The BioFire BCID2 Panel tests for 43 targets associated with bloodstream infections including bacteria, yeast, and specific antibiotic resistance genes. The overall sensitivity and specificity of the test are both >= 99%. This test can provide a result approximately one hour after the blood culture turns positive. The goal is to enhance antimicrobial stewardship by shortening the time to optimal therapy, whether this be de-escalating or escalating antimicrobials in response to the result. The table below contains generalized recommendations for optimal therapy in response to a BioFire BCID2 Panel result. Deviations from these recommendations may be warranted in the case of polymicrobial or concomitant infections or patients with a history of infection caused by multidrug resistant organisms (MDRO).

Please also consider ID consultation for the management of any bacteremia, but especially with Staphylococcus aureus, yeast, MDRO history, or reported allergies to recommended therapies. Some antibiotics below are restricted to ID approval (indicated by *).

As of April 2026 this test is not fully validated and not all results are able to be displayed in the chart. Consultation with the microbiology lab and ID team is recommended.

BCID2 Panel Result	Recommended Therapies with local antibiogram data (if applicable)
GRAM-NEGATIVE BACTERIA WITH GRAM-NEGATIVE RESISTANCE GENE DETECTED	
CTX-M extended spectrum beta-lactamase	Meropenem IV* Ertapenem IV*
***IMP* carbapenemase**	Contact ID ASAP for recommendations
***KPC* carbapenemase**	Contact ID ASAP for recommendations
***NDM* carbapenemase**	Contact ID ASAP for recommendations
***OXA-48-like* carbapenemase**	Contact ID ASAP for recommendations
***mcr-1* colistin resistance gene**	If patient is on colisin then contact ID ASAP for alternative recommendations
***VIM* carbapenemase**	Contact ID ASAP for recommendations
GRAM-NEGATIVE BACTERIA WITHOUT GRAM-NEGATIVE RESISTANCE GENE DETECTED	
Acinetobacter calcoaceticus-baumannii complex	Ampicillin-sulbactam IV (100% sensitivity) Meropenem IV* (100% sensitivity)
Bacteroides fragilis ^Since Bacteroides fragilis is commonly seen as part of a polymicrobial intra-abdominal infection additional coverage is likely warranted as metronidazole is only active against anaerobes and parasites	Ampicillin-sulbactam IV Cefoxitin IV Piperacillin-tazobactam IV Ertapenem IV* Meropenem IV* Moxifloxacin IV/PO* Metronidazole IV/PO^
Enterobacter cloacae complex	Cefepime IV* (96% sensitivity) Ciprofloxacin IV/PO* (97% sensitivity) Meropenem IV* (99% sensitivity)
Escherichia coli	Ceftriaxone IV (91% sensitivity) Piperacillin/tazobactam IV (96% sensitivity) Ertapenem IV* (100% sensitivity) Meropenem IV* (100% sensitivity)
Haemophilus influenzae	Ampicillin/sulbactam IV

	Cefuroxime IV/PO Ceftriaxone IV
Klebsiella (Enterobacter) aerogenes	Cefepime IV* (100% sensitivity) Ciprofloxacin IV/PO* (93% sensitivity) Ertapenem IV* (94% sensitivity) Meropenem IV* (100% sensitivity)
Klebsiella oxytoca	Ceftriaxone IV (92% sensitivity) Cefepime IV (98% sensitivity) Piperacillin/tazobactam (94% sensitivity) Ciprofloxacin IV/PO* (98% sensitivity)
Klebsiella pneumoniae group	Ceftriaxone IV (90% sensitivity) Ampicillin/sulbactam IV (83% sensitivity) Piperacillin/tazobactam (87% sensitivity) Ciprofloxacin IV/PO* (84% sensitivity) Ertapenem IV* (100% sensitivity) Meropenem IV* (100% sensitivity)
Neisseria meningitidis	Ceftriaxone IV
Proteus species	Ceftriaxone (90% sensitivity) Ampicillin/sulbactam IV (87% sensitivity) Piperacillin/tazobactam (99% sensitivity)
Pseudomonas aeruginosa	Piperacillin/tazobactam (91% sensitivity) Cefepime IV* (95% sensitivity) Meropenem IV* (91% sensitivity) Ciprofloxacin IV/PO* (90% sensitivity) Tobramycin IV (98% sensitivity)^ ^Consider use for double-coverage in critically ill patients with risk factors for multidrug resistant infections. Contact ID pharmacist for dosing recommendations.
Salmonella	Azithromycin IV/PO Ciprofloxacin IV/PO* Ceftriaxone IV
Serratia marcescens	Ceftriaxone IV (85% sensitivity) Cefepime IV (100% sensitivity) Piperacillin/tazobactam Ciprofloxacin IV/PO* (95% sensitivity) Ertapenem IV* (92% sensitivity) Meropenem IV* (100% sensitivity
Stenotrophomonas maltophila	Trimethoprim/sulfamethoxazole Levofloxacin IV/PO* Cefiderocol IV*

GRAM-POSITIVE BACTERIA		
Enterococcus faecalis ^Since Enterococcus is commonly seen as part of a polymicrobial intra-abdominal infection additional coverage is likely warranted. Enterococcus that is ampicillin-sensitive is also sensitive to Ampicillin/sulbactam IV and Piperacillin/tazobactam IV.	***vanA/vanB* negative** (98% of isolates)	***vanA/vanB* positive** (2% of isolates)
	Ampicillin IV^ (100% sensitivity) Vancomycin IV^ (97% sensitivity) (PCN allergy) Linezolid IV/PO (98%)	Ampicillin IV^ (98% sensitivity) Linezolid IV/PO*^ (98% sensitivity) (PCN allergy) Daptomycin IV*^ (69% sensitivity) (PCN allergy)
Enterococcus faecium ^Since Enterococcus is commonly seen as part of a polymicrobial intra-abdominal infection additional coverage is likely warranted. Enterococcus that is ampicillin-sensitive is also sensitive to Ampicillin/sulbactam IV and Piperacillin/tazobactam IV.	***vanA/vanB* negative** (33% of isolates)	***vanA/vanB* positive** (67% of isolates)
	Vancomycin IV^ (100% sensitivity)	Linezolid IV/PO*^ (93% sensitivity) Daptomycin IV*^
Listeria monocytogenes	Ampicillin IV Meropenem IV* Trimethoprim/sulfamethoxazole IV/PO	
Staphylococcus aureus	***mecA/C* and MREJ negative** (65% of isolates)	***mecA/C* and/or MREJ positive** (35% of isolates)
	Cefazolin IV (100% sensitivity) Nafcillin IV* (100% sensitivity)	Vancomycin IV (100% sensitivity) Daptomycin IV* (100% sensitivity) Linezolid IV/PO* (100% sensitivity)
Staphylococcus epidermidis^ ^The majority of blood cultures growing Staphylococcus epidermidis are from skin flora contamination. Clinically correlate for infection prior to starting antibiotics. Risk factors for true bacteremia include prosthetic material, central lines, and neutropenia.	***mecA/C* negative** (52% of isolates)	***mecA/C* positive** (48% of isolates)
	Cefazolin IV (100% sensitivity) Nafcillin IV* (100% sensitivity)	Vancomycin IV (100% sensitivity) Daptomycin IV* (100% sensitivity) Linezolid IV/PO* (100% sensitivity)
Staphylococcus lugdunensis	***mecA/C* negative**	***mecA/C* positive**
	Cefazolin IV (100% sensitivity) Nafcillin IV* (100% sensitivity)	Vancomycin IV (100% sensitivity) Daptomycin IV* (100% sensitivity) Linezolid IV/PO* (100% sensitivity)
Streptococcus agalactiae	Ampicillin IV Ceftriaxone IV Vancomycin IV	

Streptococcus pneumoniae ^If CNS infection is suspected it is recommended to do a beta-lactam AND vancomycin or linezolid until MIC testing available	Penicillin IV Cefazolin IV Ceftriaxone IV^ Vancomycin IV^ Linezolid IV/PO*
Streptococcus pyogenes	Penicillin IV Cefazolin IV Ceftriaxone IV Vancomycin IV Linezolid IV/PO

YEAST	
Candida albicans	Contact ID ASAP for recommendations Micafungin IV*
Candida auris	
Candida glabrata	
Candida krusei	
Candida parapsilosis	
Candida tropicalis	
Crypotoccus neoformans/gattii	Contact ID ASAP for recommendations